Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/09/2026
PI Name:	Thompson, Rachel Ph.D.		
Project Title:	Musculoskeletal Biomechanics Model Verification and Validation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	KBR/NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/01/2025	**End Date:**	02/28/2027
No. of Post Docs:		**No. of PhD Degrees:**	3
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	December 2025 Update: Per the PI, Lauren Hickox, Ph.D., has been added to the project team as a Co-Investigator (Ed., 1/9/26).		
COI Name (Institution):	Yates, Keegan Ph.D. (KBR/NASA Johnson Space Center) Hickox, Lauren (Aegis Aerospace/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Injury prediction and prevention in domains analogous to spaceflight dynamic events commonly rely on the use of anthropomorphic test devices (ATD), also known as crash test dummies and/or human volunteer tests. Because of the large number of potential impact conditions, and the limitations of testing apparatuses, these physical tests are typically augmented with simulations. Dynamic events are often simulated with explicit non-linear finite element analysis (FEA), but these FEA models are not designed for extended events, and running the simulations for longer durations comes at increased computational cost and probability of model instability. A lunar landing could be an extended event with additional motion occurring before or after touchdown (e.g., reaction control system thrusters firing before touchdown or landing without simultaneous contact of all landing legs). In automotive crash, pre-crash events such as braking or evasive maneuvers provide a similar issue. These extended events can instead be simulated with different types of models such as the OpenSim musculoskeletal biomechanics model. These models can predict the motion of the crewmember during extended events, and injuries can be predicted via the joint forces and torques and/or via swapping in a detailed FEA model immediately prior to a complex event, such as a head impact to a vehicle structure. This work is primarily aimed at addressing the DL-301 gap because it will help enable better prediction of injury by mission-phase and vehicle design. It also partially addresses gap DL-201 because it aims at better understanding injury risk as a function of vehicle loads.
Rationale for HRP Directed Research:	The research is highly constrained in that it is focused on defining how loads that a crewmember may experience during unique events (such as a lunar landing or surface operations) would correspond to a certain risk of injury. The nature of this directed research is further constrained since it is expected to involve the discussion of proprietary information related to provider vehicles. In-house personnel have performed research in this area and have working relationships and close proximity to subject matter experts, which will make the effort more feasible, acceptable, and as cost-efficient as possible.
Research Impact/Earth Benefits:	This work is primarily aimed at addressing the DL-301 gap because it will help enable better prediction of injury by mission-phase and vehicle design. It also partially addresses gap DL-201 because it aims at better understanding injury risk as a function of vehicle loads.
Task Progress:	The musculoskeletal modeling framework used in this study is implemented in OpenSim and represents a full-body, three-dimensional human model, intended to capture the biomechanical response of a crewmember during dynamic landing scenarios. The model integrates established, peer-reviewed component models for the lower extremities, thoraco-lumbar and cervical spine, with targeted modifications to support dynamic loading conditions relevant to spaceflight operations. Model development and validation are ongoing. Aim 1 was successfully completed, expanding the model's capabilities to include an intravehicular activity (IVA) suit configuration. Experimental data were collected to support incremental tuning and validation of the musculoskeletal model for standing landing scenarios. Experimental data were collected to support progressive tuning and validation of the model through a two-phase testing approach. The first experimental phase, completed in late 2025, involved controlled perturbations and jump landing tasks performed by multiple participants. Single-direction surface translations at varying magnitudes and awareness conditions were applied. Whole-body motion data and ground reaction forces were collected and have been pre-processed for subsequent biomechanical analysis. Future work includes a second experimental phase incorporating multi-directional perturbations using a motion platform, along with continued comparison of model-predicted joint loads to publicly available reference datasets to support verification and validation.
Bibliography Type:	Description: (Last Updated:)